

PLK1, A Potential Target for Cancer Therapy



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Abstract

Polo-like kinase 1 (*PLK1*) plays an important role in the initiation, maintenance, and completion of mitosis. Dysfunction of *PLK1* may promote cancerous transformation and drive its progression. *PLK1* overexpression has been found in a variety of human cancers and was associated with poor prognoses in cancers. Many studies have showed that inhibition of *PLK1* could lead to death of cancer cells by interfering with multiple stages of mitosis. Thus, *PLK1* is expected to be a potential target for cancer therapy. In this article, we examined *PLK1*'s structural characteristics, its regulatory roles in cell mitosis, *PLK1* expression, and its association with survival prognoses of cancer patients in a wide variety of cancer types, *PLK1* interaction networks, and *PLK1* inhibitors under investigation. Finally, we discussed the key issues in the development of *PLK1*-targeted cancer therapy.

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Introduction

Polo-like kinases (*PLKs*) are a family of serine/threonine protein kinases which are widespread in eukaryotic cells [1]. Human *PLK* family includes five members: *PLK1*, *PLK2*, *PLK3*, *PLK4*, and *PLK5*. Among them, *PLK1* was the most investigated [2]. *PLK1* plays multiple roles in the cell cycle: controls mitotic entry and the G2/M checkpoint, coordinates the centrosome and cell cycles, regulates spindle assembly and chromosome segregation, exerts multiple functions at the spindle midzone and during abscission, facilitates DNA replication, and is involved in cytokinesis and meiosis [3]. *PLK1* is essential for precisely regulating the cell division and maintaining genome stability in mitosis, spindle assembly, and DNA damage response [2,4]. Previous studies have shown that *PLK1* is highly expressed in most of human cancers, and its overexpression is associated with poor prognosis in cancer patients [5–7]. Several reports have shown that blocking the expression of *PLK1* by antibody, RNA interference (RNAi), or kinase inhibitors can effectively inhibit the proliferation of tumor cells and induce apoptosis of tumor cells [8,9]. Thus, it has been suggested that *PLK1* could be an attractive target for cancer therapy [10]. In this article, we reviewed *PLK1*'s functions in cell cycle progression, its roles in human cancers, and the development of potential inhibitors targeting *PLK1* in cancer treatment.

The Structure Characteristics of *PLK1*

As the other members in *PLK* family, *PLK1* protein involves a highly conserved N-terminal kinase catalytic domain (harboring 252 amino

acids), C-terminal polo-box domain (harboring 60–70 amino acids), and the connecting region in the middle. The N-terminal kinase domain is a Ser/Thr kinase domain with a T-loop whose phosphorylation is directly related to the kinase activity of *PLK1* [11]. The polo-box domain is a notable feature of *PLK* family. It is located in the C-terminal, each of which contains two polo-box structures, and a flexible structure is in the middle (Figure 1). The crystal structure of *PLK1* shows that the polo-box domain is similar to two clips, and the phosphopeptide is clamped in the middle [12]. Differing from the other members of *PLK* family, *PLK4* has only one polo box in the C-terminal polo-box domain [11].

PLK1 protein can bind to phosphopeptide of certain proteins through the polo-box domain. When it is recruited to a particular cell position by interacting with different phosphopeptide, its kinase domain is released. As a result, different proteins or the different sites of the same protein can be phosphorylated. In a normal condition, the

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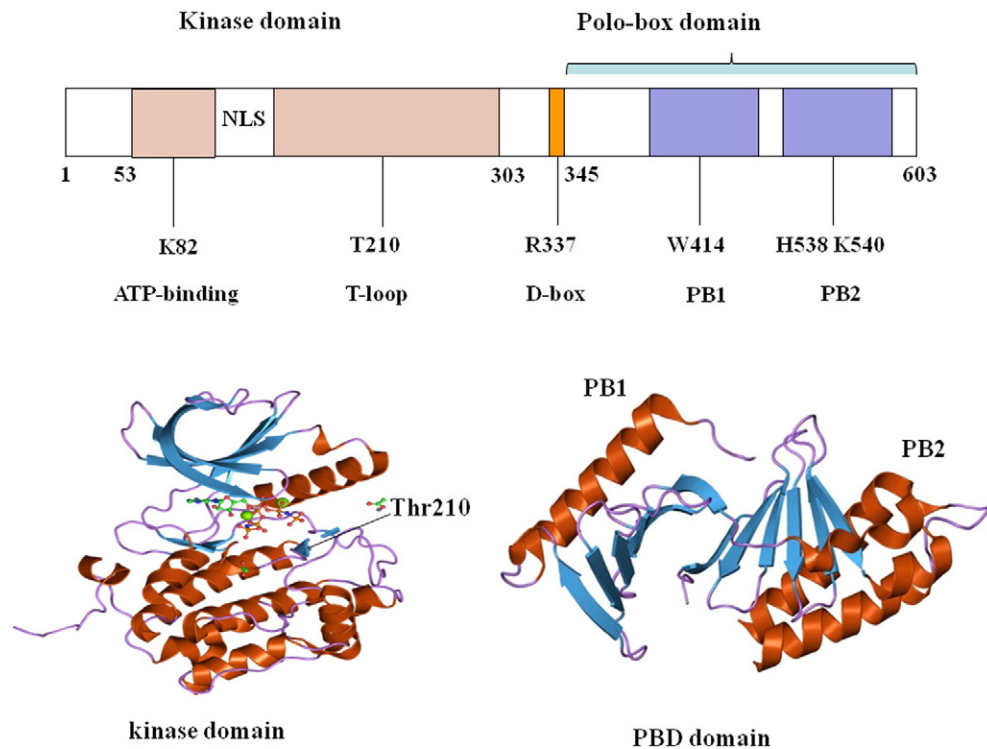


Figure 1. The domain structure of *PLK1*. PB1: polo-box 1; PB2: polo-box 2; PBD: polo-box domain.

polo-box domain always combines with the N-terminal kinase domain to inhibit the phosphorylation of T210 in the kinase, thereby inhibiting the kinase activity of *PLK1*. *PLK1* is activated when the polo-box domain binds to its ligand, and separates with the T-loop of kinase domain [13].

***PLK1* and Mitosis**

In eukaryotic organisms, cell cycle progression is regulated by proteolysis and phosphorylation, and the precise regulation is necessary for the replication of the genetic information in offspring. Any mistakes in the mitotic or DNA replication process may result in apoptosis or mutation to form tumors. *PLK1* expression is elevated in actively proliferating cells and is significantly different among different stages of the cell cycle [14]. The expression of *PLK1* is cell cycle dependent, usually gathers in the centrosome of the spindle poles in early period of mitosis, and then migrates gradually from spindle poles to the equatorial plate after entering into middle and late period of mitosis. At the end of mitosis, *PLK1* gathers in the midbody. Therefore, *PLK1* expression is barely detectable in G1 and S phase, gradually increases in G2 phase, and peaks in M phase [14]. After the completion of cell division, *PLK1* expression would get a sharp decline and then move into the next loop of cell cycles (Figure 2).

PLK1 is a key regulator of mitosis initiation. It can drive the transformation of G2/M phase by controlling the activity of the *CDK1/Cyclin B* complex which is necessary for cells' transition from the G2 phase into the M phase [15]. *PLK1* regulates cytoplasmic separation and membrane formation in mitosis telophase via phosphorylating mitotic kinesin-like protein I [16]. *PLK1* also regulates cytokinesis [17]. In late mitosis, *PLK1* protein is gradually inactivated because of the protein hydrolysis and finally enters into a quiescent state when mitosis ends.

***PLK1* and Human Cancer**

***PLK1* Expression in Cancer**

Carcinogenesis depends both on the activation of proto-oncogenes and on the deactivation of tumor suppressor genes [18]. Oncogenes and tumor suppressor genes are mostly related to cell cycle regulation, and dysregulation of the cell cycle is the main cause of cancer [19]. Since *PLK1* was found to be highly expressed in primary tumor tissues more than two decades ago [20], its role as an oncogene has been identified by many studies [21,22]. A number of studies have revealed that *PLK1* is overexpressed in cancers compared with normal controls in

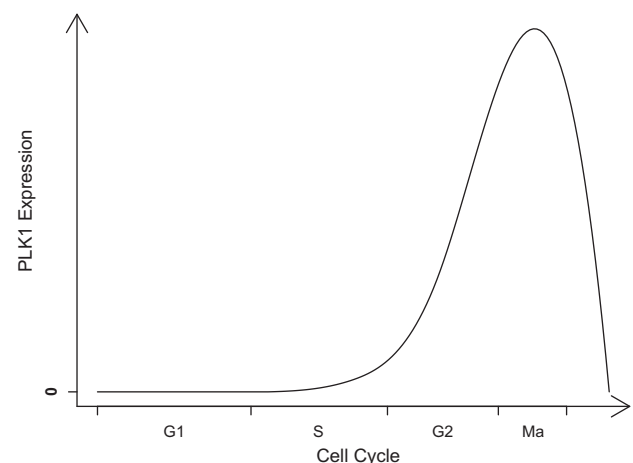


Figure 2. The tendency for *PLK1* expression in different phases of mitosis.

various types of human cancers such as glioma [23], thyroid carcinoma [24], head and neck squamous cell carcinoma [25], melanoma [26], colorectal cancers [27], esophageal carcinoma [28], ovarian carcinoma [29], breast cancer [30], and prostate cancer [31].

To examine the expression of *PLK1* in various types of human cancers, we downloaded the RNA-Seq gene expression (Level 3), gene somatic mutation (Level 2), and clinical data for all of the 33 cancer types from The Cancer Genome Atlas (TCGA) data portal (<https://gdc-portal.nci.nih.gov/>). We compared *PLK1* expression between cancers and normal tissue in 19 cancer types (14 cancer types were excluded from the analysis because of their small numbers or lack of normal samples) and found that *PLK1* has significantly higher expression levels in cancers than in normal tissue in 18 of the 19 cancer types (Student's *t* test, *P* value < .05) (Table 1). Besides the TCGA data, *PLK1* gene and protein expression has been reported to be elevated in a wide variety of human cancers compared with normal tissue in a number of previous studies and to be associated with poor prognoses of cancers (Table 2). These results confirm that the overexpression of *PLK1* gene and protein is a common feature of human cancers.

PLK1 Expression and Prognosis of Cancer

We compared the overall survival (OS) time between *PLK1* higher-expression-level and *PLK1* lower-expression-level cancers in 25 cancer types (8 cancer types were excluded from the analysis because of very few samples having both *PLK1* expression and OS data). *PLK1* higher-expression-level and lower-expression-level cancer patients were determined by the median values of *PLK1* expression. If the *PLK1* expression level in a patient was higher than the median value, the patient was classified as *PLK1* higher expression level; otherwise, patient was classified as *PLK1* lower expression level. Kaplan-Meier survival curves (Figure 3) show that patients with higher expression levels of *PLK1* have significantly worse OS prognoses than those with lower expression levels of *PLK1* in 10 cancer types: adrenocortical carcinoma (ACC), BLCA, BRCA, KIRC, KIRP, brain lower-grade glioma (LGG), LUAD, PAAD, skin cutaneous melanoma (SKCM), and UCEC. In the other cancer types, both groups of cancer patients show no significant OS time differences (log-rank test, *P* value < .05). We also compared the disease-free survival (DFS) time between *PLK1* higher-expression-level and *PLK1* lower-expression-level cancers in 26 cancer types (7 cancer types were excluded from the analysis because of very few

Table 2. Overexpression of *PLK1* mRNA and Protein in Human Cancers Reported in Previous Studies

Cancer Type	Reference
Lung cancer	[32,33]
Breast cancer	[30,34–36]
Melanoma	[26,37]
Renal cancer	[38,39]
Head and neck cancer	[25,40,41]
Hepatocellular carcinoma	[42–44]
Endometrial carcinoma	[45]
Colorectal cancer	[6,27,46,47]
Gastric carcinoma	[48,49]
Esophageal carcinoma	[50,51]
Bladder urothelial carcinoma	[52]
Prostate cancer	[31]
Cholangiocarcinoma	[53]
Glioblastoma	[54,55]
Glioma	[23,56]
Ovarian cancer	[29,57]
Pancreatic cancer	[58–60]
Thyroid cancer	[24]

samples having both *PLK1* expression and DFS data). Kaplan-Meier survival curves (Figure 4) show that patients with higher expression levels of *PLK1* have significantly worse DFS prognoses than those with lower expression levels of *PLK1* in seven cancer types: ACC, KIRC, KIRP, LGG, LUAD, SKCM, and uveal melanoma (UVM). In the other cancer types, both groups of cancer patients show no significant DFS time differences (log-rank test, *P* value < .05). These results confirm that overexpression of *PLK1* leads to poor clinical outcomes in cancer [5,7].

Moreover, we compared *PLK1* expression levels between early and late stage of cancer patients in 11 TCGA cancer types which have relatively complete records of the clinical phenotype. The 11 cancer types include ACC, BLCA, BRCA, cervical squamous cell carcinoma and endocervical adenocarcinoma (CESC), CHOL, COAD, lymphoid neoplasm diffuse large B-cell lymphoma (DLBC), ESCA, GBM, HNSC, and KICH. We found that *PLK1* has significantly higher expression levels in late stage of cancers than in early stage of cancers in four cancer types: ACC (Student's *t* test, *P* value = 2×10^{-5} , fold change = 3.3), BRCA (*P* value = .03, fold change = 1.9), KIRC (*P* value = 1.7×10^{-6} , fold change = 1.6), and KIRP (*P* value = .0005, fold change = 1.6). In the other seven cancer types, *PLK1* expression levels have no significant differences between early and late stage of cancers, although the mean expression levels of *PLK1* are higher in late stage of cancers than in early stage of cancers in six of the seven cancer types (except COAD). These results indicate that *PLK1* upregulation may be involved in cancer progression and invasion. This is in line with the results of previous studies [60–62].

Interactions between PLK1 and Other Molecules

PLK1 interacts with a number of gene products (proteins) (Figure 5, generated by the BioGRID [63]). For example, *RSF1* directly binds to *PLK1* to play an important role in *PLK1* deposition and function at mitotic kinetochores [64]; *PLK1* phosphorylates *PAX3-FOXO1* to stabilize the protein and has been proposed as a rational target for treating alveolar rhabdomyosarcoma [65]; *CLIP-170* recruits *PLK1* to kinetochores during early mitosis for chromosome alignment [66]. Among the interactive partners with *PLK1*, tumor suppressor genes are noteworthy considering the oncogenic role of *PLK1* in human cancers. The tumor suppressor *p53* acts as the “guardian of the genome” and plays an important role in antiproliferation [67]. *TP53* mutations and

Table 1. Comparison of *PLK1* Expression between Cancers and Normal Tissue

Cancer Type	Full Name	<i>P</i> Value	Fold Change*
LUSC	Lung squamous cell carcinoma	7.41E-157	20.8
BRCA	Breast invasive carcinoma	5.88E-126	11.3
LUAD	Lung adenocarcinoma	1.18E-63	9.7
KIRC	Kidney renal clear cell carcinoma	2.33E-55	6.1
HNSC	Head and neck squamous cell carcinoma	6.52E-50	4.2
LIHC	Liver hepatocellular carcinoma	3.59E-40	11.7
UCEC	Uterine corpus endometrial carcinoma	1.96E-36	21.3
COAD	Colon adenocarcinoma	5.97E-33	2.5
STAD	Stomach adenocarcinoma	8.45E-27	4.8
ESCA	Esophageal carcinoma	9.52E-27	10.2
BLCA	Bladder urothelial carcinoma	4.96E-26	9.1
PRAD	Prostate adenocarcinoma	1.29E-22	3.3
KIRP	Kidney renal papillary cell carcinoma	6.76E-22	4.7
CHOL	Cholangiocarcinoma	6.97E-14	24.3
GBM	Glioblastoma multiforme	5.63E-12	12.4
KICH	Kidney chromophobe	1.63E-06	3.3
READ	Rectum adenocarcinoma	1.06E-05	2.3
PAAD	Pancreatic adenocarcinoma	0.04	2.2

* Mean *PLK1* expression in cancers/mean *PLK1* expression in normal tissue.

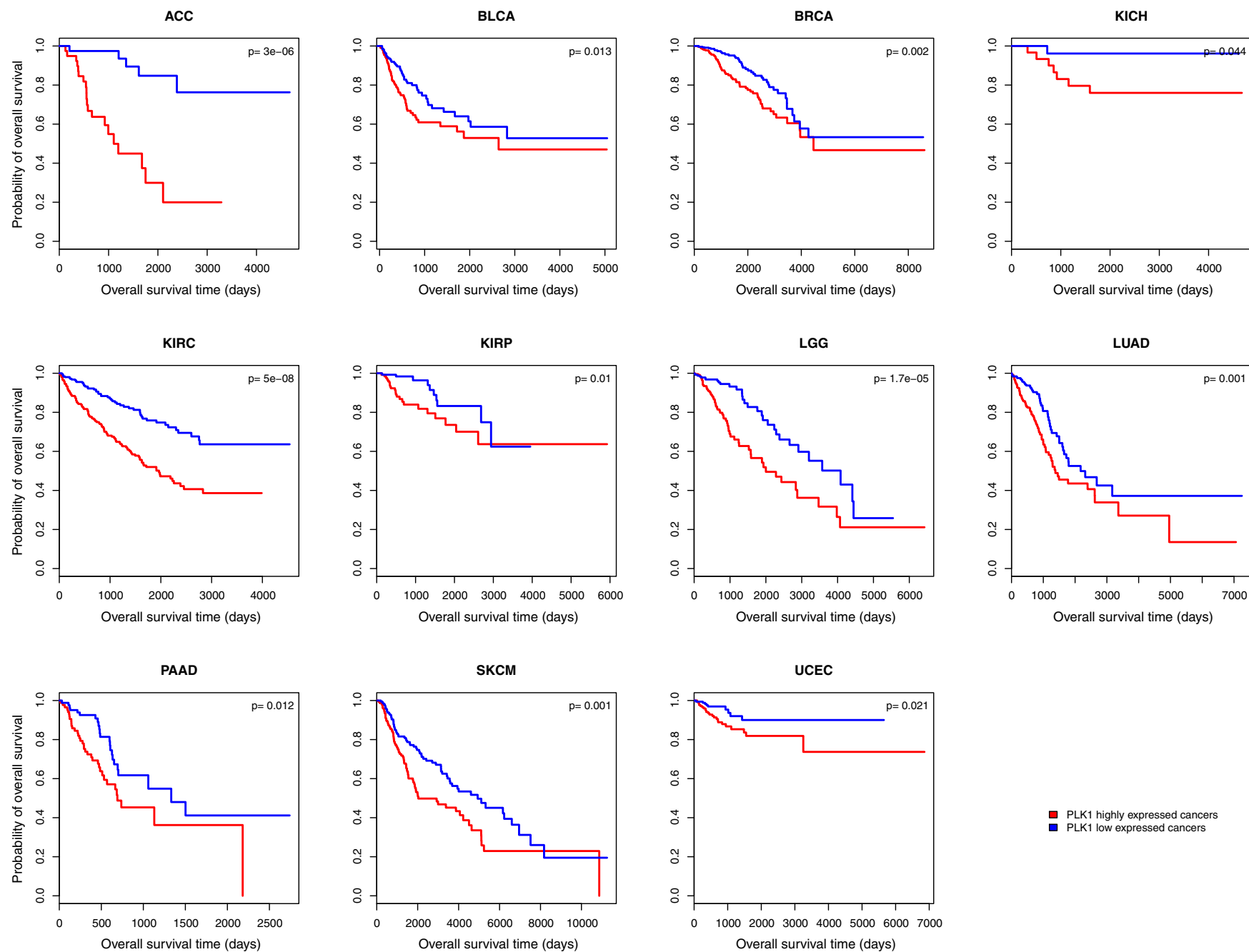


Figure 3. Kaplan-Meier survival curves show significant overall survival (OS) time differences between *PLK1* higher-expression-level and *PLK1* lower-expression-level cancer patients (log-rank test, P value $< .05$).

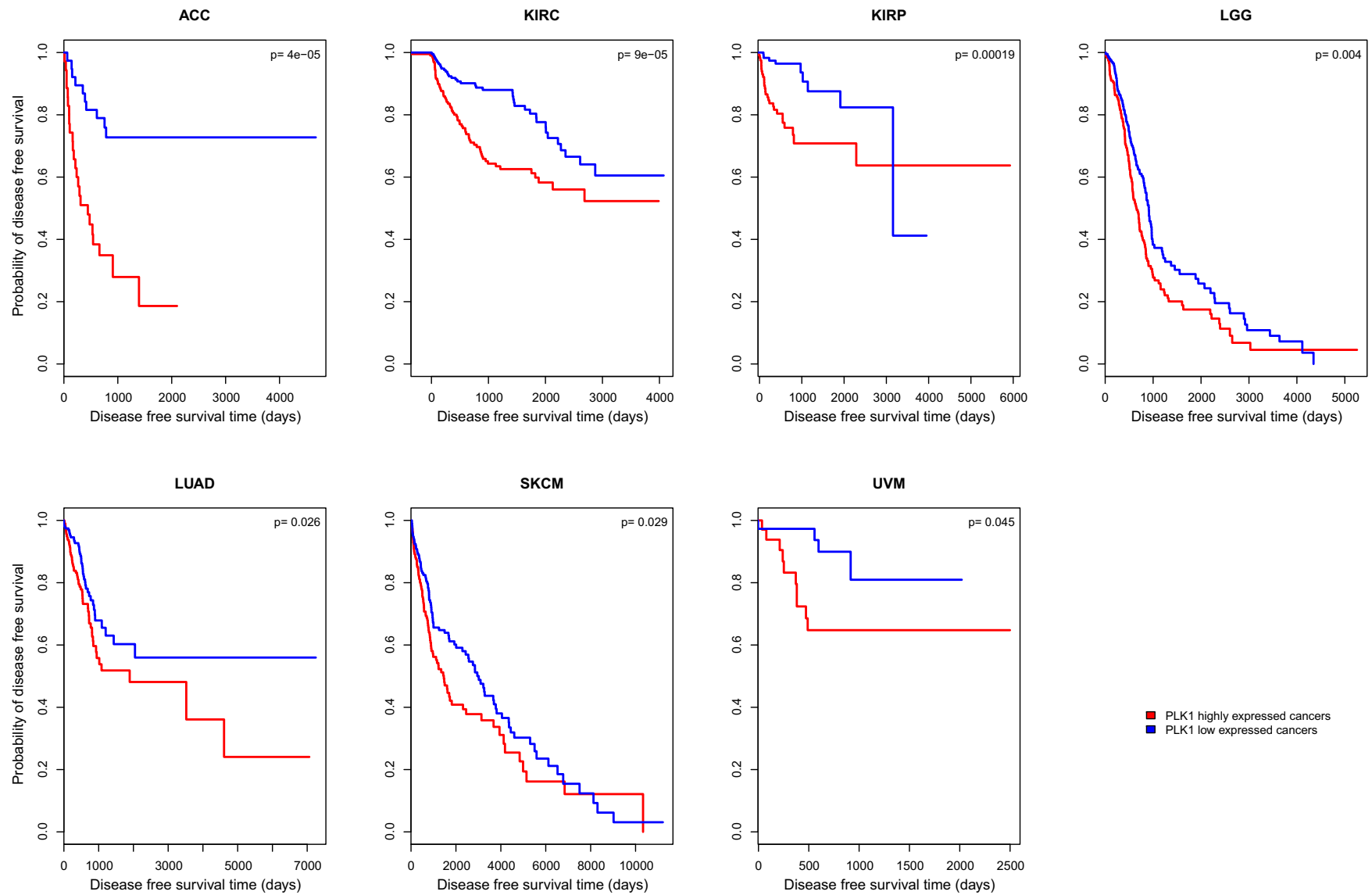


Figure 4. Kaplan-Meier survival curves show significant disease-free survival (DFS) time differences between *PLK1* higher-expression-level and *PLK1* lower-expression-level cancer patients (log-rank test, P value < .05).

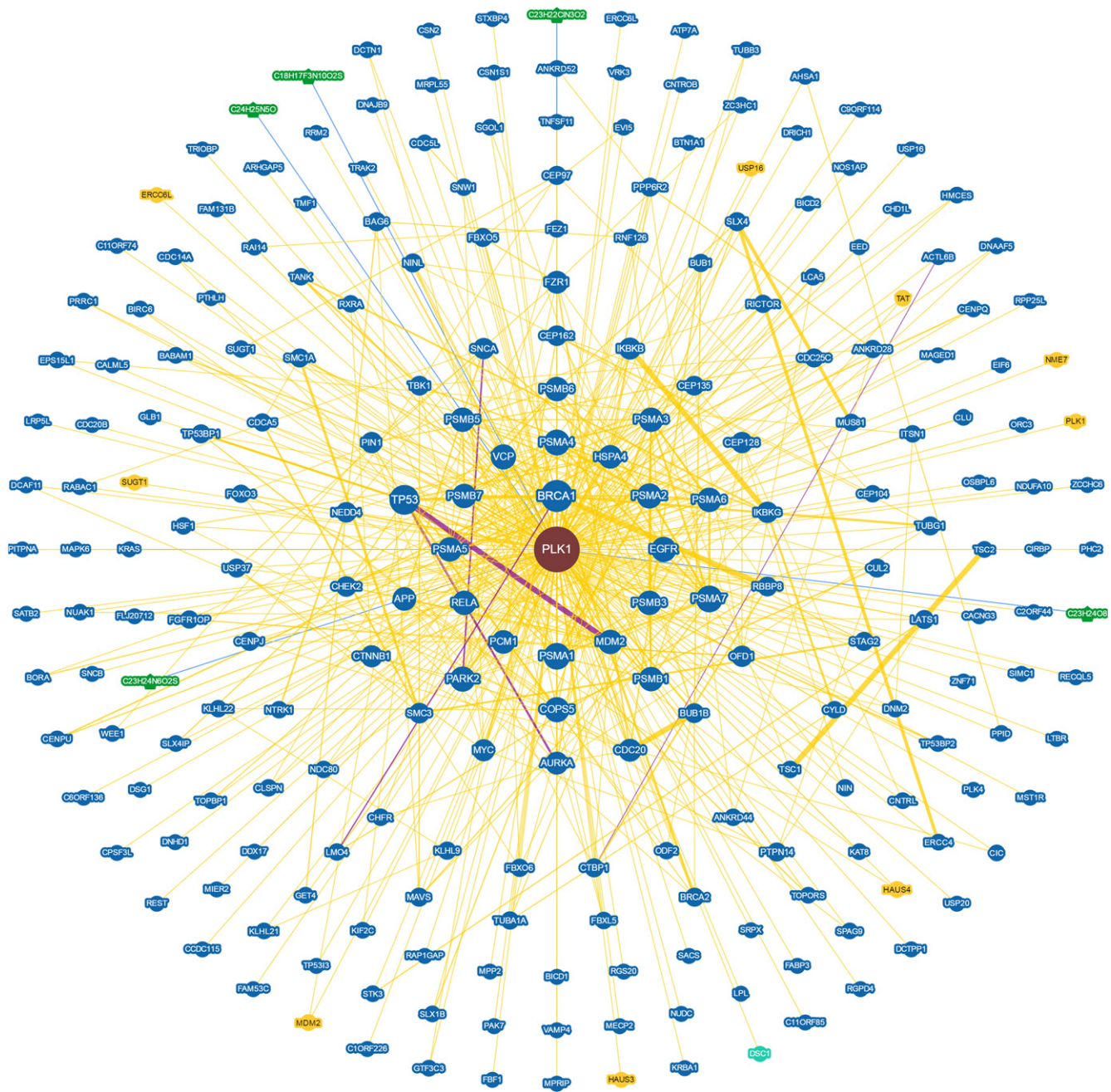


Figure 5. *PLK1* interaction networks.

dysfunction occur in more than half of all human cancer cases [68]. Previous studies have shown that *PLK1* can bind to the sequence-specific DNA-binding domain of *p53* and inhibit the *p53*-dependent transcriptional activation as well as proapoptotic activity by physical interaction and phosphorylation [69]. *P53* in turn can repress expression of *PLK1* [70]. By analyses of the TCGA datasets, we found that *PLK1* expression levels are significantly higher in *TP53*-mutated cancers than in *TP53*-wild-type cancers in 17 of the 29 cancer types (4 cancer types were excluded from the analysis because of their small numbers of *TP53*-mutated samples), as shown in Table 3. These results suggest that *p53* may repress *PLK1* expression, and once *TP53* mutations result in loss of the *p53* transcriptional repression function, *PLK1* would have elevated expression in *TP53*-mutated cancers compared with *TP53*-wild-type cancers.

When we focused on the aforementioned 19 cancer types each of which has a sufficient number of normal control samples, we found that in 11 of the 19 cancer types, *PLK1* expression follows this pattern: *TP53*-mutated cancers > *TP53*-wild-type cancers > normal controls. The 11 cancer types include BRCA, LUAD, UCEC, BLCA, LIHC, PRAD, STAD, KIRC, LUSC, KICH, and CHOL (Figure 6). This pattern indicates that *PLK1* has an elevated expression level in cancers relative to normal tissue and further has an elevated expression level in *TP53*-mutated cancers than in *TP53*-wild-type cancers, suggesting that the interaction between the oncogene *PLK1* and the tumor suppressor gene *TP53* may play an important role in carcinogenesis.

Some other tumor suppressors such as *CHK2* [71], *BRCA1/2* [72–75], *ATM* [76] and *ATR* [77], *BUB1B* [78,79], *CYLD* [80],

Table 3. Comparison of *PLK1* Expression between *TP53*-Mutated and *TP53*-Wild-Type Cancers

Cancer Type	Full Name	P Value	Fold Change*
BRCA	Breast invasive carcinoma	6.45E-52	2.5
LUAD	Lung adenocarcinoma	5.14E-23	2.1
UCEC	Uterine corpus endometrial carcinoma	1.10E-15	2.2
BLCA	Bladder urothelial carcinoma	2.44E-14	1.8
LIHC	Liver hepatocellular carcinoma	6.72E-13	2.5
PRAD	Prostate adenocarcinoma	1.12E-07	1.8
STAD	Stomach adenocarcinoma	1.57E-07	1.5
ACC	Adrenocortical carcinoma	2.85E-07	5.4
PAAD	Pancreatic adenocarcinoma	6.15E-06	1.7
KIRC	Kidney renal clear cell carcinoma	0.0004	2.3
SKCM	Skin cutaneous melanoma	0.0004	1.4
SARC	Sarcoma	0.001	1.5
LUSC	Lung squamous cell carcinoma	0.01	1.3
CESC	Cervical squamous cell carcinoma and endocervical adenocarcinoma	0.02	1.4
KICH	Kidney chromophobe	0.03	1.9
DLBC	Lymphoid neoplasm diffuse large B-cell Lymphoma	0.03	1.8
CHOL	Cholangiocarcinoma	0.05	2.4

* Mean *PLK1* expression in *TP53*-mutated cancers/mean *PLK1* expression in *TP53*-wild-type cancers.

REST [81], and *TSC1/2* [82,83] all have been shown to interact with *PLK1*. The imbalance between these interactions for regulating proliferation could be a leading cause of cancer.

In addition to the tumor suppressors, *PLK1* interacts with other classes of genes or proteins such as kinases: *AURKA* [84,85], *BUB1* [86,87], and *WEE1* [88]; oncogenes: *MDM2* [89] and *FOXO3* [90]; and transcriptional factors: *HSF1* [91–93], *RELA* [94], *TRIOBP* [95], and *FOXMI* [62,96–98], etc. These interactions may be essential for *PLK1* to play an important role in regulation of the cell cycle, and dysfunction of the interactions could be associated with cancer and other diseases.

PLK1 and Tumor Sensitivity to Chemotherapy and Radiotherapy

Drug resistance of cancer cells is one of the main reasons for the failure of chemotherapy in clinic [99,100]. A number of studies have revealed that targeting *PLK1* may be a novel approach for overcoming drug resistance in cancer chemotherapy. For example, Jimeno et al. confirmed that *PLK1* mediated the resistance to gemcitabine of pancreatic cancer cells [101]; Tyagi et al. reported that *PLK1* silencing could enhance the sensitivity to cisplatin in human *TP53*-mutated epidermal squamous carcinoma cells by upregulating *p73a* [102]; Gleixner et al. found that BI2536, a small molecule inhibitor of *PLK1*, could enhance the inhibition of imatinib and nilotinib on chronic myeloid leukemia cell growth [103].

In addition, many studies have suggested that *PLK1* is potentially important for radiation sensitizer. For example, Rodel et al. showed that *PLK1* silencing could enhance the sensitivity of rectal cancer to radiotherapy [47]; Gerster et al. revealed that targeting *PLK1* enhanced radiation efficacy for HNSC [40]; Harris et al. found that using BI2536 before radiotherapy could enhance radiotherapy sensitivity in medulloblastoma cells [104]. Thus, the inhibition of *PLK1* may overcome drug resistance in cancer chemotherapy and enhance sensitivity of cancer radiotherapy.

PLK1 as a Target for Cancer Therapy

PLK1 could be a new therapeutic target for cancer because *PLK1* knockout can decrease cancer cell survival, induce apoptosis, and increase the sensitivity to chemotherapy drugs, whereas it has little effect on normal cells [9,105–107]. A number of studies have shown that inhibiting *PLK1*

expression or function by RNAi or small molecule inhibitors was effective in control of cancer cell proliferation [38,50,108,109].

RNAi or small interfering RNA (siRNA) technology is used to inhibit certain gene expression by human intervention. Spänkuch-Schmitt et al. showed that siRNAs targeting *PLK1* reduced cancer cell proliferation, whereas they had almost no effect on human mammary epithelial cells [110]. McCarroll et al. targeted *PLK1* by RNA-interfering nanoparticle-7 that reduced non-small cell lung cancer cell proliferation in a mouse model [22]. A number of studies have suggested that targeting *PLK1* by siRNA could be a viable approach to cancer therapy [9,60,111]. However, as the synthesis of antisense oligonucleotides is susceptible to ribozymes' attack and RNAi has security and stability issues, small molecule inhibitors may be a better option in targeting *PLK1* for cancer therapy than RNAi. *PLK1* structure provides two targets: kinase domain of N-terminal and polo-box domain of C-terminal. Thus, we can divide *PLK1* inhibitors into two classes, *ATP*-competitive inhibitors and non-*ATP*-competitive inhibitors, based on their different action mechanisms [112] (Table 4).

ATP-competitive inhibitors target the deep groove in the kinase *ATP* binding domain [112]. BI2536 is a representative *ATP*-competitive inhibitor with strong selective inhibition on *PLK1* [103]. It can inhibit the proliferation of various cancer cell lines from different tissue sources by blocking cancer cells in the metaphase of mitosis and leading to apoptosis [130,131]. Volasertib (BI6727) is another promising *PLK1* inhibitor. Several preclinical experiments have demonstrated that BI6727 is highly efficacious in inducing tumor regression [116,132–134]. As a result, this agent has recently been awarded the “Breakthrough Therapy Status” by the Food and Drug Administration for its significant benefit in treating acute myeloid leukemia patients [135]. However, because of the high conservation of *ATP* binding domains of different kinases and the frequent mutation in *ATP* binding sites, cancer patients always develop resistance to *ATP*-competitive inhibitors [136]. In addition, the *ATP*-competitive inhibitors can often also act on other kinases and therefore are not specific for *PLK1*.

In contrast, the polo-box domain is specific to *PLKs* and therefore could be a more suitable target for development of selective *PLK1* inhibitors. Poloxin, thymoquinone, and purpurogallin are selective *PLK1* inhibitors targeting the polo-box domain of *PLK1* [137]. Poloxin and thymoquinone can block the correct orientation of *PLK1*, thereby preventing the mitosis of cancer cells [126]. A recent study showed that Poloxin-2, an optimized analogue of poloxin, has significantly improved potency and selectivity over poloxin in inducing mitotic arrest and apoptosis in cultured human cancer cells [128].

However, so far, the small molecule inhibitors of *PLK1* including BI2536 have not achieved a satisfactory therapeutic effect in clinical trials [138]. One main reason is the dose-limiting toxicities of *PLK1* inhibitors [10]. A recent study revealed that reduced efficacy of the *PLK1* inhibitor BI2536 on progressive hepatocellular carcinoma was due to low intratumoral drug levels [139].

Concluding Remarks

PLK1 is a key regulator of the cell cycle and an important oncogene in cancer initiation, progression, and drug resistance. Its overexpression is a common feature of human cancers (Tables 1 and 2) and is an important marker for prognosis in cancer (Figures 3 and 4). Inhibition of *PLK1* expression could reverse the drug resistance of cancer cells and increase sensitivity to radiotherapy and chemotherapy. Thus, *PLK1* could be a promising target for cancer treatment. Particularly, Wang and Simon have proposed that *PLK1* is a promising target for treating *TP53*-mutated

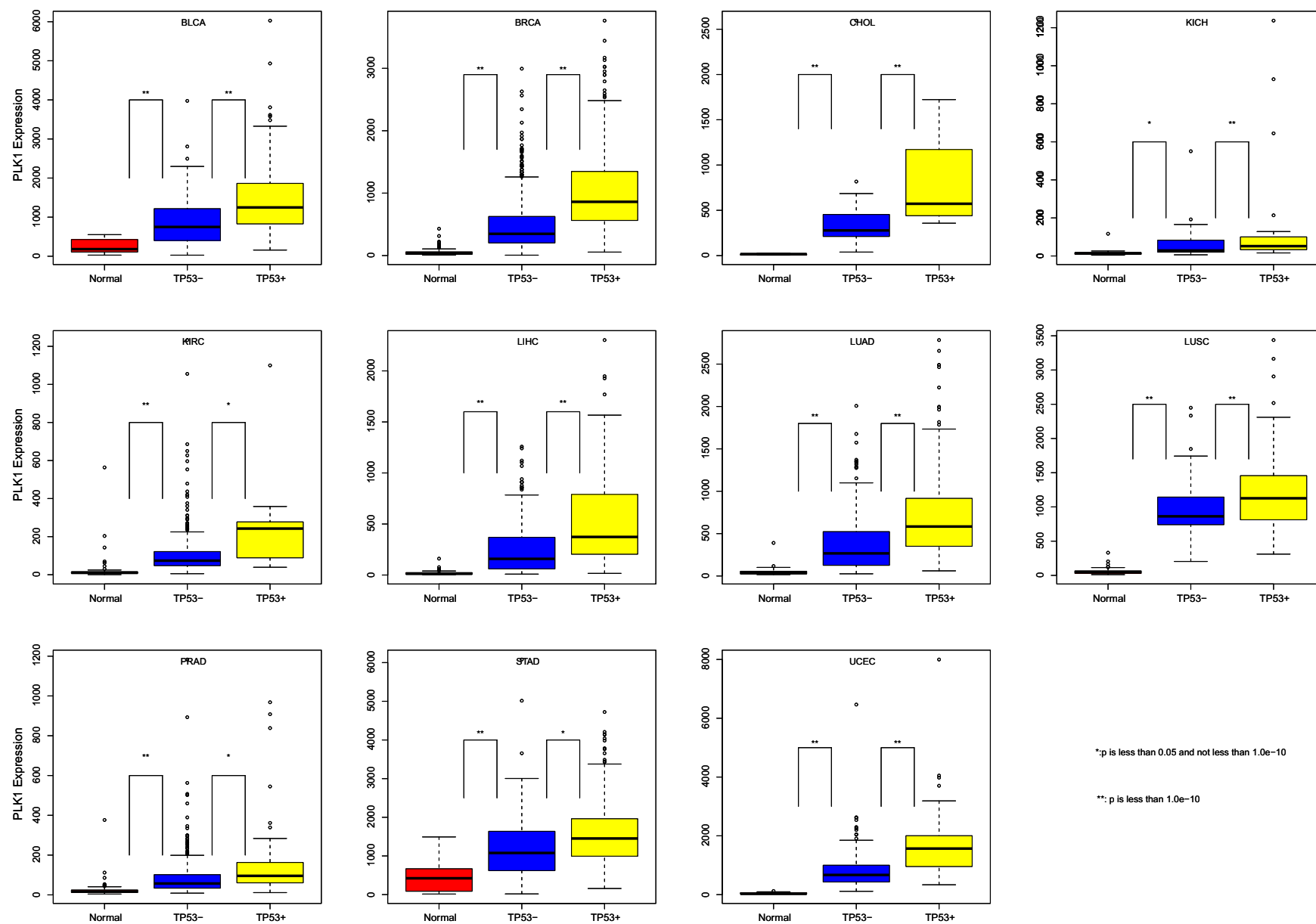


Figure 6. *PLK1* gene expression level pattern: *TP53*-mutated cancers > *TP53*-wild-type cancers > normal controls, in 11 cancer types. *TP53*+: *TP53*-mutated cancers; *TP53*-: *TP53*-wild-type cancers.

Table 4. A List of *PLK1* Inhibitors

Inhibitor	Status	Reference	Company or Lab	Class
BI2536	Experimental	[113]	Boehringer Ingelheim	ATP-competitive
GSK461364	Experimental	[114,115]	GlaxoSmithKline	ATP-competitive
Volasertib (BI6727)	Experimental	[115,116]	Boehringer Ingelheim	ATP-competitive
ZK-thiazolidinone	Experimental	[115,117]	Bayer Schering Pharmacy	ATP-competitive
Rigosertib (ON01910)	Experimental	[118]	Onconova Therapeutics Inc.	Non-ATP-competitive
Cyclapolin 9	Experimental	[119,120]	Cyclacel	ATP-competitive
GW 843682X	Experimental	[121]	GlaxoSmithKline	ATP-competitive
SBE 13 hydrochloride	Experimental	[122,123]	Institute of Organic Chemistry & Chemical Biology, Goethe-University	ATP-competitive
TAK960 hydrochloride	Experimental	[124,125]	Takeda Pharmaceutical Company	ATP-competitive
Poloxin	Experimental	[126,127]	Max Planck Institute of Biochemistry and Munich Center for Integrated Protein Science	Non-ATP-competitive
Poloxin-2	Experimental	[128]	Institute of Organic Chemistry, University of Leipzig	Non-ATP-competitive
RO3280	Experimental	[129]	Hoffmann-La Roche	ATP-competitive

cancers because of its potential synthetic lethality relationship with *TP53* [140]. However, to translate the cancer biology of *PLK1* into clinical application, we need to resolve some important issues such as the following: Is the overexpression of *PLK1* in cancer the cause of cancer or just a consequence of cancer cells' proliferation [141]? What is the key interactive network of *PLK1* underlying the carcinogenesis? What is the true relationship between *p53* and *PLK1* in cancer? How can we develop effective *PLK1* inhibitors or improve the clinical efficacy of current *PLK1* inhibitors? We believe that the solution to these issues would bring significant progress in cancer treatment by targeting *PLK1* and its related network or pathway.

Competing Interests

The authors declare that they have no competing interests.

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References

[1] Golsteyn RM, et al (1996). The family of polo-like kinases. *Prog Cell Cycle Res* 2, 107–114.

[2] Lens SM, Voest EE, and Medema RH (2010). Shared and separate functions of polo-like kinases and aurora kinases in cancer. *Nat Rev Cancer* 10(12), 825–841.

[3] van de Weerd BC and Medema RH (2006). Polo-like kinases: a team in control of the division. *Cell Cycle* 5(8), 853–864.

[4] Mundt KE, et al (1997). On the regulation and function of human polo-like kinase 1 (PLK1): effects of overexpression on cell cycle progression. *Biochem Biophys Res Commun* 239(2), 377–385.

[5] Ramani P, et al (2015). High levels of polo-like kinase 1 and phosphorylated translationally controlled tumor protein indicate poor prognosis in neuroblastomas. *J Neurooncol* 125(1), 103–111.

[6] Tut TG, et al (2015). Upregulated polo-like kinase 1 expression correlates with inferior survival outcomes in rectal cancer. *PLoS One* 10(6), e0129313.

[7] Zhang R, et al (2015). Misregulation of polo-like protein kinase 1, P53 and P21WAF1 in epithelial ovarian cancer suggests poor prognosis. *Oncol Rep* 33(3), 1235–1242.

[8] de Oliveira JC, et al (2012). In vitro PLK1 inhibition by BI 2536 decreases proliferation and induces cell-cycle arrest in melanoma cells. *J Drugs Dermatol* 11(5), 587–592.

[9] Bu Y, et al (2008). Silencing of polo-like kinase (Plk) 1 via siRNA causes inhibition of growth and induction of apoptosis in human esophageal cancer cells. *Oncology* 74(3–4), 198–206.

[10] Degenhardt Y and Lampkin T (2010). Targeting polo-like kinase in cancer therapy. *Clin Cancer Res* 16(2), 384–389.

[11] de Carcer G, Manning G, and Malumbres M (2011). From Plk1 to Plk5: functional evolution of polo-like kinases. *Cell Cycle* 10(14), 2255–2262.

[12] Cheng KY, et al (2003). The crystal structure of the human polo-like kinase-1 polo box domain and its phospho-peptide complex. *EMBO J* 22(21), 5757–5768.

[13] Lowery DM, Lim D, and Yaffe MB (2005). Structure and function of polo-like kinases. *Oncogene* 24(2), 248–259.

[14] Schmucker S and Sumara I (2014). Molecular dynamics of PLK1 during mitosis. *Mol Cell Oncol* 1(2), e954507.

[15] Gavet O and Pines J (2010). Activation of cyclin B1-Cdk1 synchronizes events in the nucleus and the cytoplasm at mitosis. *J Cell Biol* 189(2), 247–259.

[16] Liu X, et al (2004). Molecular interactions of polo-like-kinase 1 with the mitotic kinesin-like protein CHO1/MKLP-1. *J Cell Sci* 117(Pt 15), 3233–3246.

[17] Takaki T, et al (2008). Polo-like kinase 1 reaches beyond mitosis—cytokinesis, DNA damage response, and development. *Curr Opin Cell Biol* 20(6), 650–660.

[18] Kopnin BP (2000). Targets of oncogenes and tumor suppressors: key for understanding basic mechanisms of carcinogenesis. *Biochemistry (Mosc)* 65(1), 2–27.

[19] Yang L, et al (2015). Mutations of p53 and KRAS activate NF-kappaB to promote chemoresistance and tumorigenesis via dysregulation of cell cycle and suppression of apoptosis in lung cancer cells. *Cancer Lett* 357(2), 520–526.

[20] Holtrich U, et al (1994). Induction and down-regulation of PLK, a human serine/threonine kinase expressed in proliferating cells and tumors. *Proc Natl Acad Sci U S A* 91(5), 1736–1740.

[21] Gray Jr PJ, et al (2004). Identification of human polo-like kinase 1 as a potential therapeutic target in pancreatic cancer. *Mol Cancer Ther* 3(5), 641–646.

[22] McCarroll JA, et al (2015). Therapeutic targeting of polo-like kinase 1 using RNA-interfering nanoparticles (iNOPs) for the treatment of non-small cell lung cancer. *Oncotarget* 6(14), 12020–12034.

[23] Cheng MW, et al (2012). Clinicopathological significance of polo-like kinase 1 (PLK1) expression in human malignant glioma. *Acta Histochem* 114(5), 503–509.

[24] Ito Y, et al (2004). Polo-like kinase 1 overexpression is an early event in the progression of papillary carcinoma. *Br J Cancer* 90(2), 414–418.

[25] Knecht R, et al (1999). Prognostic significance of polo-like kinase (PLK) expression in squamous cell carcinomas of the head and neck. *Cancer Res* 59(12), 2794–2797.

[26] Kneisel L, et al (2002). Expression of polo-like kinase (PLK1) in thin melanomas: a novel marker of metastatic disease. *J Cutan Pathol* 29(6), 354–358.

[27] Takahashi T, et al (2003). Polo-like kinase 1 (PLK1) is overexpressed in primary colorectal cancers. *Cancer Sci* 94(2), 148–152.

[28] Tokumitsu Y, et al (1999). Prognostic significance of polo-like kinase expression in esophageal carcinoma. *Int J Oncol* 15(4), 687–692.

[29] Weichert W, et al (2004). Polo-like kinase 1 overexpression is a prognostic factor in ovarian carcinoma. *Br J Cancer* 90(4), 815–821.

[30] Weichert W, et al (2005). Polo-like kinase isoforms in breast cancer: expression patterns and prognostic implications. *Virchows Arch* 446(4), 442–450.

[31] Weichert W, et al (2004). Polo-like kinase 1 is overexpressed in prostate cancer and linked to higher tumor grades. *Prostate* 60(3), 240–245.

[32] Wolf G, et al (1997). Prognostic significance of polo-like kinase (PLK) expression in non-small cell lung cancer. *Oncogene* 14(5), 543–549.

[33] Ferrarotto R, et al (2016). Epithelial-mesenchymal transition predicts polo-like kinase 1 inhibitor-mediated apoptosis in non-small cell lung cancer. *Clin Cancer Res* 22(7), 1674–1686.

[34] Donizy P, et al (2016). Augmented expression of polo-like kinase 1 is a strong predictor of shorter cancer-specific overall survival in early stage breast cancer at 15-year follow-up. *Oncol Lett* 12(3), 1667–1674.

- [35] Wolf G, et al (2000). Polo-like kinase: a novel marker of proliferation: correlation with estrogen-receptor expression in human breast cancer. *Pathol Res Pract* **196**(11), 753–759.
- [36] King SI, et al (2012). Immunohistochemical detection of polo-like kinase-1 (PLK1) in primary breast cancer is associated with TP53 mutation and poor clinical outcome. *Breast Cancer Res* **14**(2), R40.
- [37] Strebhardt K, et al (2000). Prognostic value of pololike kinase expression in melanomas. *JAMA* **283**(4), 479–480.
- [38] Ding Y, et al (2011). Combined gene expression profiling and RNAi screening in clear cell renal cell carcinoma identify PLK1 and other therapeutic kinase targets. *Cancer Res* **71**(15), 5225–5234.
- [39] Zhang GJ, Zhang Z, and Liu ZG (2013). Polo-like kinase 1 is overexpressed in renal cancer and participates in the proliferation and invasion of renal cancer cells. *Tumor Biol* **34**(3), 1887–1894.
- [40] Gerster K, et al (2010). Targeting polo-like kinase 1 enhances radiation efficacy for head-and-neck squamous cell carcinoma. *Int J Radiat Oncol Biol Phys* **77**(1), 253–260.
- [41] Knecht R, Oberhauser C, and Strebhardt K (2000). PLK (polo-like kinase), a new prognostic marker for oropharyngeal carcinomas. *Int J Cancer* **89**(6), 535–536.
- [42] He ZL, et al (2009). Overexpression of polo-like kinase1 predicts a poor prognosis in hepatocellular carcinoma patients. *World J Gastroenterol* **15**(33), 4177–4182.
- [43] Mok WC, et al (2012). Polo-like kinase 1, a new therapeutic target in hepatocellular carcinoma. *World J Gastroenterol* **18**(27), 3527–3536.
- [44] Sun W, et al (2014). High expression of polo-like kinase 1 is associated with early development of hepatocellular carcinoma. *Int J Genomics*.
- [45] Takai N, et al (2001). Polo-like kinase (PLK) expression in endometrial carcinoma. *Cancer Lett* **169**(1), 41–49.
- [46] Weichert W, et al (2005). Polo-like kinase 1 expression is a prognostic factor in human colon cancer. *World J Gastroenterol* **11**(36), 5644–5650.
- [47] Rodel F, et al (2010). Polo-like kinase 1 as predictive marker and therapeutic target for radiotherapy in rectal cancer. *Am J Pathol* **177**(2), 918–929.
- [48] Weichert W, et al (2006). Expression patterns of polo-like kinase 1 in human gastric cancer. *Cancer Sci* **97**(4), 271–276.
- [49] Dibb M, et al (2015). FOXM1 and polo-like kinase 1 are co-ordinately overexpressed in patients with gastric adenocarcinomas. *BMC Res Notes* **8**, 676.
- [50] Zhao CL, et al (2015). Downregulation of PLK1 by RNAi attenuates the tumorigenicity of esophageal squamous cell carcinoma cells via promoting apoptosis and inhibiting angiogenesis. *Neoplasia* **62**(5), 748–755.
- [51] Feng YB, et al (2009). Overexpression of PLK1 is associated with poor survival by inhibiting apoptosis via enhancement of survivin level in esophageal squamous cell carcinoma. *Int J Cancer* **124**(3), 578–588.
- [52] Zhang Z, Zhang GJ, and Kong CZ (2013). High expression of polo-like kinase 1 is associated with the metastasis and recurrence in urothelial carcinoma of bladder. *Urol Oncol* **31**(7), 1222–1230.
- [53] Thrum S, et al (2011). Polo-like kinase 1 inhibition as a new therapeutic modality in therapy of cholangiocarcinoma. *Anticancer Res* **31**(10), 3289–3299.
- [54] Lee C, et al (2012). Polo-like kinase 1 inhibition kills glioblastoma multiforme brain tumor cells in part through loss of SOX2 and delays tumor progression in mice. *Stem Cells* **30**(6), 1064–1075.
- [55] Danovi D, et al (2013). A high-content small molecule screen identifies sensitivity of glioblastoma stem cells to inhibition of polo-like kinase 1. *PLoS One* **8**(10), e77053.
- [56] Dietzmann K, et al (2001). Increased human polo-like kinase-1 expression in gliomas. *J Neurooncol* **53**(1), 1–11.
- [57] Takai N, et al (2001). Expression of polo-like kinase in ovarian cancer is associated with histological grade and clinical stage. *Cancer Lett* **164**(1), 41–49.
- [58] Weichert W, et al (2005). Overexpression of polo-like kinase 1 is a common and early event in pancreatic cancer. *Pancreatology* **5**(2–3), 259–265.
- [59] Zhang C, et al (2012). Validation of polo-like kinase 1 as a therapeutic target in pancreatic cancer cells. *Cancer Biol Ther* **13**(12), 1214–1220.
- [60] Mahajan UM, et al (2016). Tumour-specific delivery of siRNA-coupled superparamagnetic iron oxide nanoparticles, targeted against PLK1, stops progression of pancreatic cancer. *Gut*.
- [61] Rizki A, Mott JD, and Bissell MJ (2007). Polo-like kinase 1 is involved in invasion through extracellular matrix. *Cancer Res* **67**(23), 11106–11110.
- [62] Zhang Z, Zhang G, and Kong C (2016). FOXM1 participates in PLK1-regulated cell cycle progression in renal cell cancer cells. *Oncol Lett* **11**(4), 2685–2691.
- [63] Stark C, et al (2006). BioGRID: a general repository for interaction datasets. *Nucleic Acids Res* **34**(Database issue), D535–D539.
- [64] Lee HS, et al (2015). The chromatin remodeller RSF1 is essential for PLK1 deposition and function at mitotic kinetochores. *Nat Commun* **6**.
- [65] Thalhammer V, et al (2015). PLK1 phosphorylates PAX3-FOXO1, the inhibition of which triggers regression of alveolar rhabdomyosarcoma. *Cancer Res* **75**(1), 98–110.
- [66] Amin MA, et al (2014). CLIP-170 recruits PLK1 to kinetochores during early mitosis for chromosome alignment. *J Cell Sci* **127**(Pt 13), 2818–2824.
- [67] Vogelstein B, Lane D, and Levine AJ (2000). Surfing the p53 network. *Nature* **408**(6810), 307–310.
- [68] Hollstein M, et al (1991). p53 mutations in human cancers. *Science* **253**(5015), 49–53.
- [69] Ando K, et al (2004). Polo-like kinase 1 (Plk1) inhibits p53 function by physical interaction and phosphorylation. *J Biol Chem* **279**(24), 25549–25561.
- [70] McKenzie L, et al (2010). p53-dependent repression of polo-like kinase-1 (PLK1). *Cell Cycle* **9**(20), 4200–4212.
- [71] Tsvetkov L, et al (2003). Polo-like kinase 1 and Chk2 interact and co-localize to centrosomes and the midbody. *J Biol Chem* **278**(10), 8468–8475.
- [72] Zou J, et al (2013). BRCA1 downregulates the kinase activity of polo-like kinase 1 in response to replication stress. *Cell Cycle* **12**(14), 2255–2265.
- [73] Chabaliere-Taste C, et al (2016). Polo-like kinase 1 mediates BRCA1 phosphorylation and recruitment at DNA double-strand breaks. *Oncotarget* **7**(3), 2269–2283.
- [74] Lee M, Daniels MJ, and Venkitaraman AR (2004). Phosphorylation of BRCA2 by the polo-like kinase Plk1 is regulated by DNA damage and mitotic progression. *Oncogene* **23**(4), 865–872.
- [75] Lin HR, et al (2003). M phase-specific phosphorylation of BRCA2 by polo-like kinase 1 correlates with the dissociation of the BRCA2-P/CAF complex. *J Biol Chem* **278**(38), 35979–35987.
- [76] Lee HJ, Hwang HI, and Jang YJ (2010). Mitotic DNA damage response: polo-like kinase-1 is dephosphorylated through ATM-Chk1 pathway. *Cell Cycle* **9**(12), 2389–2398.
- [77] Deming PB, et al (2002). ATR enforces the topoisomerase II-dependent G2 checkpoint through inhibition of Plk1 kinase. *J Biol Chem* **277**(39), 36832–36838.
- [78] Elowe S, et al (2007). Tension-sensitive Plk1 phosphorylation on BubR1 regulates the stability of kinetochore microtubule interactions. *Genes Dev* **21**(17), 2205–2219.
- [79] Izumi H, et al (2009). BubR1 localizes to centrosomes and suppresses centrosome amplification via regulating Plk1 activity in interphase cells. *Oncogene* **28**(31), 2806–2820.
- [80] Stegmeier F, et al (2007). The tumor suppressor CYLD regulates entry into mitosis. *Proc Natl Acad Sci U S A* **104**(21), 8869–8874.
- [81] Karlin KL, et al (2014). The oncogenic STP axis promotes triple-negative breast cancer via degradation of the REST tumor suppressor. *Cell Rep* **9**(4), 1318–1332.
- [82] Astrinidis A, Senapedis W, and Henske EP (2006). Hamartin, the tuberous sclerosis complex 1 gene product, interacts with polo-like kinase 1 in a phosphorylation-dependent manner. *Hum Mol Genet* **15**(2), 287–297.
- [83] Rosner M, et al (2008). The tuberous sclerosis gene products hamartin and tuberin are multifunctional proteins with a wide spectrum of interacting partners. *Mutat Res* **658**(3), 234–246.
- [84] Asteriti IA, De Mattia F, and Guarguaglini G (2015). Cross-talk between AURKA and Plk1 in mitotic entry and spindle assembly. *Front Oncol* **5**, 283.
- [85] Macurek L, et al (2008). Polo-like kinase-1 is activated by aurora A to promote checkpoint recovery. *Nature* **455**(7209), 119–123.
- [86] Jia L, Li B, and Yu H (2016). The Bub1-Plk1 kinase complex promotes spindle checkpoint signalling through Cdc20 phosphorylation. *Nat Commun* **7**, 10818.
- [87] Qi W, Tang Z, and Yu H (2006). Phosphorylation- and polo-box-dependent binding of Plk1 to Bub1 is required for the kinetochore localization of Plk1. *Mol Biol Cell* **17**(8), 3705–3716.
- [88] De Witt Hamer PC, et al (2011). WEE1 kinase targeting combined with DNA-damaging cancer therapy catalyzes mitotic catastrophe. *Clin Cancer Res* **17**(13), 4200–4207.
- [89] Dias SS, et al (2009). Polo-like kinase-1 phosphorylates MDM2 at Ser260 and stimulates MDM2-mediated p53 turnover. *FEBS Lett* **583**(22), 3543–3548.
- [90] Bucur O, et al (2014). PLK1 is a binding partner and a negative regulator of FOXO3 tumor suppressor. *Discoveries (Craiova)* **2**(2).
- [91] Kim SA, et al (2010). The antitumor effect of PLK1 and HSF1 double knockdown on human oral carcinoma cells. *Int J Oncol* **36**(4), 867–872.
- [92] Lee YJ, et al (2008). HSF1 as a mitotic regulator: phosphorylation of HSF1 by Plk1 is essential for mitotic progression. *Cancer Res* **68**(18), 7550–7560.
- [93] Yoon JH, Kim SA, and Ahn SG (2008). Antitumor effect of PLK1 and HSF1 double knockdown in human squamous cell carcinoma. *FEBS J* **275**, 294.
- [94] Kuo HP, et al (2013). Epigenetic roles of MLL oncoproteins are dependent on NF-kappaB. *Cancer Cell* **24**(4), 423–437.

- [95] Zhu Y, et al (2012). Phosphorylation of Tara by Plk1 is essential for faithful chromosome segregation in mitosis. *Exp Cell Res* **318**(18), 2344–2352.
- [96] Dibb M, et al (2012). The FOXM1-PLK1 axis is commonly upregulated in oesophageal adenocarcinoma. *Br J Cancer* **107**(10), 1766–1775.
- [97] Fu Z, et al (2008). Plk1-dependent phosphorylation of FoxM1 regulates a transcriptional programme required for mitotic progression. *Nat Cell Biol* **10**(9), 1076–1082.
- [98] Wang R, et al (2013). The expression of Nek7, FoxM1, and Plk1 in gallbladder cancer and their relationships to clinicopathologic features and survival. *Clin Transl Oncol* **15**(8), 626–632.
- [99] Gillet JP and Gottesman MM (2010). Mechanisms of multidrug resistance in cancer. *Methods Mol Biol* **596**, 47–76.
- [100] Hu X and Zhang Z (2016). Understanding the genetic mechanisms of cancer drug resistance using genomic approaches. *Trends Genet* **32**(2), 127–137.
- [101] Jimeno A, et al (2010). A fine-needle aspirate-based vulnerability assay identifies polo-like kinase 1 as a mediator of gemcitabine resistance in pancreatic cancer. *Mol Cancer Ther* **9**(2), 311–318.
- [102] Tyagi S, et al (2010). Polo-like kinase1 (Plk1) knockdown enhances cisplatin chemosensitivity via up-regulation of p73alpha in p53 mutant human epidermoid squamous carcinoma cells. *Biochem Pharmacol* **80**(9), 1326–1334.
- [103] Gleixner KV, et al (2010). Polo-like kinase 1 (Plk1) as a novel drug target in chronic myeloid leukemia: overriding imatinib resistance with the Plk1 inhibitor BI 2536. *Cancer Res* **70**(4), 1513–1523.
- [104] Harris PS, et al (2012). Polo-like kinase 1 (PLK1) inhibition suppresses cell growth and enhances radiation sensitivity in medulloblastoma cells. *BMC Cancer* **12**, 80.
- [105] Weiss L and Efferth T (2012). Polo-like kinase 1 as target for cancer therapy. *Exp Hematol Oncol* **1**(1), 38.
- [106] Reagan-Shaw S and Ahmad N (2005). Silencing of polo-like kinase (Plk) 1 via siRNA causes induction of apoptosis and impairment of mitosis machinery in human prostate cancer cells: implications for the treatment of prostate cancer. *FASEB J* **19**(6), 611–613.
- [107] Benoit DS, et al (2010). pH-responsive polymeric sirna carriers sensitize multidrug resistant ovarian cancer cells to doxorubicin via knockdown of polo-like kinase 1. *Mol Pharm* **7**(2), 442–455.
- [108] Bowles DW, et al (2014). Phase I study of oral rigosertib (ON 01910.Na), a dual inhibitor of the PI3K and Plk1 pathways, in adult patients with advanced solid malignancies. *Clin Cancer Res* **20**(6), 1656–1665.
- [109] Chou YS, et al (2016). Cytotoxic mechanism of PLK1 inhibitor GSK461364 against osteosarcoma: mitotic arrest, apoptosis, cellular senescence, and synergistic effect with paclitaxel. *Int J Oncol* **48**(3), 1187–1194.
- [110] Spankuch-Schmitt B, et al (2002). Effect of RNA silencing of polo-like kinase-1 (PLK1) on apoptosis and spindle formation in human cancer cells. *J Natl Cancer Inst* **94**(24), 1863–1877.
- [111] Hu K, et al (2012). Small interfering RNA library screen identified polo-like kinase-1 (PLK1) as a potential therapeutic target for breast cancer that uniquely eliminates tumor-initiating cells. *Breast Cancer Res* **14**(1), R22.
- [112] Li S, Zhang Y, and Xu W (2013). Developments of polo-like kinase 1 (Plk1) inhibitors as anti-cancer agents. *Mini Rev Med Chem* **13**(14), 2014–2025.
- [113] Steegmaier M, et al (2007). BI 2536, a potent and selective inhibitor of polo-like kinase 1, inhibits tumor growth in vivo. *Curr Biol* **17**(4), 316–322.
- [114] Emmitte KA, et al (2009). Design of potent thiophene inhibitors of polo-like kinase 1 with improved solubility and reduced protein binding. *Bioorg Med Chem Lett* **19**(6), 1694–1697.
- [115] Schoffski P (2009). Polo-like kinase (PLK) inhibitors in preclinical and early clinical development in oncology. *Oncologist* **14**(6), 559–570.
- [116] Rudolph D, et al (2009). BI 6727, a polo-like kinase inhibitor with improved pharmacokinetic profile and broad antitumor activity. *Clin Cancer Res* **15**(9), 3094–3102.
- [117] Santamaria A, et al (2007). Use of the novel Plk1 inhibitor ZK-thiazolidinone to elucidate functions of Plk1 in early and late stages of mitosis. *Mol Biol Cell* **18**(10), 4024–4036.
- [118] Gumireddy K, et al (2005). ON01910, a non-ATP-competitive small molecule inhibitor of Plk1, is a potent anticancer agent. *Cancer Cell* **7**(3), 275–286.
- [119] McInnes C, et al (2006). Inhibitors of polo-like kinase reveal roles in spindle-pole maintenance. *Nat Chem Biol* **2**(11), 608–617.
- [120] Taylor P, et al (2008). Ligand discovery and virtual screening using the program LIDAEUS. *Br J Pharmacol* **153**, S55–S67.
- [121] Lansing TJ, et al (2007). In vitro biological activity of a novel small-molecule inhibitor of polo-like kinase 1. *Mol Cancer Ther* **6**(2), 450–459.
- [122] Liu X (2010). SBE13 joins the family of polo-like kinase 1 (Plk1) inhibitors. *Cell Cycle* **9**(3), 445–446.
- [123] Keppner S, et al (2009). Identification and validation of a potent type II inhibitor of inactive polo-like kinase 1. *ChemMedChem* **4**(11), 1806–1809.
- [124] Hikichi Y, et al (2012). TAK-960, a novel, orally available, selective inhibitor of polo-like kinase 1, shows broad-spectrum preclinical antitumor activity in multiple dosing regimens. *Mol Cancer Ther* **11**(3), 700–709.
- [125] Nie Z, et al (2013). Discovery of TAK-960: an orally available small molecule inhibitor of polo-like kinase 1 (PLK1). *Bioorg Med Chem Lett* **23**(12), 3662–3666.
- [126] Reindl W, et al (2008). Inhibition of polo-like kinase 1 by blocking polo-box domain-dependent protein–protein interactions. *Chem Biol* **15**(5), 459–466.
- [127] Yuan J, et al (2011). Polo-box domain inhibitor poloxin activates the spindle assembly checkpoint and inhibits tumor growth in vivo. *Am J Pathol* **179**(4), 2091–2099.
- [128] Scharow A, et al (2015). Optimized Plk1 PBD inhibitors based on poloxin induce mitotic arrest and apoptosis in tumor cells. *ACS Chem Biol* **10**(11), 2570–2579.
- [129] Chen S, et al (2012). Identification of novel, potent and selective inhibitors of Polo-like kinase 1. *Bioorg Med Chem Lett* **22**(2), 1247–1250.
- [130] Eckerdt F (2011). Polo-like kinase 1 inhibitors SBE13 and BI 2536 induce different responses in primary cells. *Cell Cycle* **10**(7), 1027–1028.
- [131] Wasch R, et al (2010). BI_2536—targeting the mitotic kinase polo-like kinase 1 (Plk1). *Recent Results Cancer Res* **184**, 215–218.
- [132] Gorlick R, et al (2014). Initial testing (stage 1) of the polo-like kinase inhibitor volasertib (BI 6727), by the Pediatric Preclinical Testing Program. *Pediatr Blood Cancer* **61**(1), 158–164.
- [133] Rudolph D, et al (2015). Efficacy and mechanism of action of volasertib, a potent and selective inhibitor of polo-like kinases, in preclinical models of acute myeloid leukemia. *J Pharmacol Exp Ther* **352**(3), 579–589.
- [134] Cholewa BD, et al (2014). Large-scale label-free comparative proteomics analysis of polo-like kinase 1 inhibition via the small-molecule inhibitor BI 6727 (volasertib) in BRAF(V600E) mutant melanoma cells. *J Proteome Res* **13**(11), 5041–5050.
- [135] Van den Bossche J, et al (2016). Spotlight on volasertib: preclinical and clinical evaluation of a promising Plk1 inhibitor. *Med Res Rev* **36**(4), 749–786.
- [136] Buschbeck M (2006). Strategies to overcome resistance to targeted protein kinase inhibitors in the treatment of cancer. *Drugs R D* **7**(2), 73–86.
- [137] Liao C, et al (2010). Exploring potential binding modes of small drug-like molecules to the polo-box domain of human polo-like kinase 1. *ACS Med Chem Lett* **1**(3), 110–114.
- [138] Gutteridge RE, et al (2016). Plk1 inhibitors in cancer therapy: from laboratory to clinics. *Mol Cancer Ther* **15**(7), 1427–1435.
- [139] Hauptenthal J, et al (2012). Reduced efficacy of the Plk1 inhibitor BI 2536 on the progression of hepatocellular carcinoma due to low intratumoral drug levels. *Neoplasia* **14**(5), 410–419.
- [140] Wang X and Simon R (2013). Identification of potential synthetic lethal genes to p53 using a computational biology approach. *BMC Med Genomics* **6**, 30.
- [141] Cholewa BD, Liu X, and Ahmad N (2013). The role of polo-like kinase 1 in carcinogenesis: cause or consequence? *Cancer Res* **73**(23), 6848–6855.